

Jev State Engine

A deterministic finite-state machine (LangGraph) that orchestrates an untrusted LLM writing code with strict Inversion of Control.

The Problem

- LLM code agents suffer from cascading hallucination.
- Unstructured loops write unconstrained diffs, introducing silent syntax breaks.
- Agents wander outside task scope and hallucinate file paths.
- Prompt-only rules fail to enforce mechanical boundaries.

The Core Idea: Inversion of Control

- "The LLM proposes, deterministic code decides."
- FSM (LangGraph) dictates execution flow, memory, and permissions.
- Worker LLM (Gemini) generates atomic changes within state constraints.
- Gatekeeper (Jev API & Tier 0) deterministically judges every diff.

End-to-End Pipeline

- 1. Ticket Submission
- 2. Investigation (Read-Only context gathering)
- 3. Planning (Grounding & Subgoal Generation)
- 4. Implementation (Isolated Git Worktree mutations)
- 5. Gating (Tier 0 Mechanical & Tier 1 Semantic checks)
- 6. Verification (Multi-ecosystem tests & Final sign-off)
- 7. Clean Fast-Forward Merge

Node: Investigation

- Reads repo to understand architecture and paths.
- Strictly Read-Only tools (`list_dir`, `grep`, `read_file`).
- Write tools are physically omitted from bindings.
- Outputs: `investigation_notes`, `investigated_directories`.

Node: Planning

- Deconstructs ticket into an ordered queue of subgoals.
- Outputs strict Pydantic Subgoal (description, scope, expects_tests).
- Grounding Guard: Scopes must reference investigated or ticket-defined files.
- Error routing: Retries on JSON/schema errors before escalating.

Node: Implementation

- Applies mutations inside an isolated Git worktree.
- Branch: jev-subgoal-<id>, Worktree: .jev-worktrees/subgoal-<id>.
- Restricted Tools: stage_file_mutation, submit_subgoal.
- Injects previous gate feedback into worker context.

Node: Gating (Two-Tier)

- Tier 0 (Mechanical): AST validation, Scope creeping checks, Test runners.
- Tier 1 (Semantic): TypeSafe AI Jev System One probability verdict.
- Independent Strike Counters (Max 3 strikes each).
- Pass -> Fast-forward merge. Fail -> Rollback worktree, increment strike, retry.

Node: Verification & Completion

- Final validation after all subgoals complete.
- Executes multi-ecosystem repository test suites.
- Final sign-off by Jev Gatekeeper using cumulative git diff.
- Completion or Escalation to Human-in-the-Loop (HITL).

The Two-Tier Gate

- Tier 0 protects against syntax errors and out-of-scope file modifications.
- Docs-Only Classifier bypasses tests for comment/markdown-only diffs.
- Tier 1 uses calibrated boolean evaluation ('Does diff meet subgoal?').
- Both tiers must pass to merge code to the main branch.

Git Worktree Isolation

- Every subgoal runs in a hidden .jev-worktrees directory.
- Protects the main repository from dirty state and half-completed features.
- Enforces atomic, fast-forward-only merges.
- Does NOT provide OS-level sandboxing (files are on the same disk).

State & Checkpointing

- Uses LangGraph State TypedDict to manage FSM transitions.
- SqliteSaver persists checkpoints to 'checkpoints.db' for run resumption.
- AST State-Guard (test_state_guard.py) ensures no undeclared keys are dropped.
- Maintains full telemetry and failure node trajectory in escalation.log.

Directory Grounding of Plans

- Real Problem: LLMs frequently hallucinate paths (e.g. src/pages vs src/app).
- Mechanical Fix: Planning node enforces scope validation against investigated directories.
- Follow-up hole: Rejects non-candidate files explicitly in tests to tighten grounding.
- Verified by
test_node_plan_rejects_new_directory_segment_when_in_notes_only_a
s_plain_word.

Read-Only Tool Gating

- Investigation node only provides `list_dir`, `grep`, `read_file`.
- Write tools are intentionally withheld at the prompt/binding level.
- Honest Limit: The tools are not physically unlinked at an OS or protocol level.
- Relies on LLM function calling limits, not true system permissions.

Multi-Ecosystem Detection

- Automatically detects and runs tests for major ecosystems.
- Supports: npm, yarn, pnpm, Go, Rust, Maven, Gradle, Pytest.
- IaC: Terraform, Ansible, Helm linting.
- Python uses fast ast.parse for instantaneous compilation/syntax gating.

Architectural Design Decisions

- RULE 0: Never assume, never hardcode. Halt on undefined constraints.
- Decision: FSM as Master. Reason: Replaces non-deterministic thick agents.
- Decision: Fail-closed on missing tests. Reason: Ensures code is actually verified.
- Decision: Docs-only classification. Reason: Avoids test framework failure on README edits.

Real Bugs & End-to-End Runs

- Note: Claims regarding 'jp-bat' disposable test site runs could not be verified.
- Git history and codebase contain no mention of 'jp-bat' or specific E2E bugs against it.
- Mechanically enforced bug fixes are covered by the comprehensive unit tests.
- This claim remains formally unverified.

Test Suite Coverage

- 220 tests collected across unit and integration levels.
- Covers FSM routing, Gatekeeper HTTP, schemas, Workspace isolation, and Test Runners.
- Excludes live LLM API behavior when skipping 'live' marked tests.
- Ensures mechanical safety and framework correctness.

Repository Map

- `src/main.py` - CLI entrypoint and wiring
- `src/jev/engine.py` - LangGraph FSM definition
- `src/jev/gatekeeper.py` - TypeSafe AI HTTP Client
- `src/jev/models.py` - Pydantic schemas
- `src/jev/workspace.py` - Git worktree lifecycle & test runners
- `tests/` - 220 Pytest files

Honest Limitations & Risks

- No OS-level sandboxing (Worktrees isolate git state only).
- No MCP-level tool gating (Tools are software-bound, not protocol isolated).
- Cross-platform testing not fully established.
- Single-worker-model dependence (Gemini).
- No License file, No CI pipeline.

Roadmap Status

- Done / Implemented: Multi-ecosystem runners, Worktree isolation, Two-Tier gating.
- Not Started: OS-level Sandboxing / Containerization.
- Not Started: MCP tool gating.
- Not Started: Cross-platform CI tests, License file.
- Not Started: User-selectable planning model.

Next Steps & Ask

- [PRESENTER PLACEHOLDER: Insert next steps, funding request, or deployment plan here]

Appendix A: Glossary

- FSM: Finite-State Machine, deterministic graph.
- Worktree: Isolated Git working directory (.jev-worktrees).
- Tier 0: Mechanical, deterministic gating (AST, tests).
- Semantic Judge: Tier 1 Jev LLM evaluating code correctness.
- Escalation: Halting automation to request Human-in-the-Loop review.

Appendix B: How to Run

- 1. Configure .env with JEV_API_KEY and GOOGLE_API_KEY.
- 2. Run: `python -m src.main "TICKET DESCRIPTION"`
- Options:
 - `--repo-dir ./target-repo`
 - `--thread-id ticket-101`
 - `--db-path checkpoints.db`

Unverified / Needs Confirmation

- The project references 'jp-bat' disposable test site and specific real bugs found in end-to-end runs.
- These claims are absent from the git history, commits, and source code.
- As a result, no verified comparison of an earlier failed run vs later clean run against jp-bat can be provided.